

Exploring A Large Archival Collection through LLM Entity Resolution

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Abstract – This paper discusses the integration of a Large Language Model (LLM) into an archival research workflow to explore entity resolution across historically sensitive datasets. The discussion centers on a current Computational Archival Science (CAS) project that seeks to match Japanese American individuals from the Sacramento, CA area across datasets extracted from census documents recorded in 1940 and 1950. The ability to identify individuals present both before and after the Japanese American World War II Internment Camps may allow further insight into socioeconomic impacts of internment, backed by data from a large archival collection. However, the challenges involved with the data present in the census records suggests that an LLM may be best suited for analysis and entity resolution.

Keywords – Computational Archival Science (CAS), Artificial Intelligence (AI), Large Language Models (LLM), Digital Humanities, entity resolution, archival research, collections as data

I. Introduction

The use of Generative Artificial Intelligence (GenAI), namely Large Language Models (LLMs), is a growing topic of exploration in the field of archival research. LLMs have been assessed as a means to support research in various ways, such as through generating survey materials, modeling topics, or writing base code. Several publications spotlighting the use of commercially available LLMs such as ChatGPT and Claude Sonnet have contributed to discussions regarding the integration of GenAI into archival and Digital Humanities project workflows. The project discussed in this paper seeks to apply such a workflow to research involving a historically sensitive, novel archival collection in order to highlight the successes and challenges of using a commercially available LLM as a computational tool in humanities research.

In February of 1942, months after the bombing of Pearl Harbor, President Franklin D. Roosevelt signed Executive Order 9066, which mandated the displacement and detention of predominantly Japanese Americans into “relocation centers,” later known as internment camps. The National Archives has recently made available a large collection of census records from the 1950s, including those specific to the Sacramento area of California, which was home to a significant Japanese American population both before and after wartime incarceration. The records were handwritten and numbered in the thousands, making traditional humanities research methods of parsing the documents by hand difficult. To better read, extract, and analyze data from such a large and complex collection, computational methods are expected to assist not only in analysis of the 1950s set of records, but also in linking data to the 1940s set of records in order to paint a comprehensive picture of subjects in the collection. The project discussed in this paper explores the use of a commercially available LLM, ChatGPT, to identify individuals who returned to Sacramento after internment by comparing lists of Japanese American residents from the 1940s records with the 1950s records. While the project is currently unfinished, the workflow established thus far offers insight into the availability of AI tools to support archival research.

II. Related Work

Previously, other tools have been used to work with the Sacramento census records that are featured in this research. Jansen and Marciano used machine learning and computer vision in optical character recognition to digitally read document texts, and were thus able to filter records of individuals whose race was identified as White (2025). This paved the way for textual data to be extracted and digital spreadsheets to be created, allowing for ease of use by future researchers. For this project, spreadsheets were filtered by the Race field to identify Japanese American subjects. Additionally, Beltran et. al. explored internment camp records through the lens of computational prosopography and tackled the question of entity resolution with fuzzy matching both manually and through a Python-created soundex (2021). This identified different approaches to matching individuals across multiple datasets within the context of the internment camps. The project discussed in this paper will build on these findings by attempting to increase entity resolution accuracy through incorporation of record fields other than name (relative matching, birth year, etc.). The inclusion of additional fields is expected to reduce the number of false positive matches and perhaps allow for more fuzzy matches for individuals whose names vary across datasets. The complexity of comparing additional fields is a challenge for which traditional or manual methods may be difficult, but for which this project hopes to address through LLM analysis.

III. Constraints and Concerns

This project uses OpenAI's ChatGPT to perform LLM analysis of the Sacramento Census datasets, as well as to match individuals across census datasets. ChatGPT in particular is a readily available and recognizable tool by researchers across fields with both technical and non-technical backgrounds, due to its efficiency for several types of tasks and for its accessibility to the public. It is also recognized by Gnanasekaran et al. (2024) as the preferred Generative AI tool for working with historically and culturally sensitive records. Researchers found that the paid tier of Azure OpenAI's ChatGPT-4o successfully provided context-rich answers to historical questions about their dataset and successfully rejected queries for use of derogatory language. Such considerations are relevant to archival humanities research that works with historically sensitive collections. The enterprise tier is used in this project to ensure security and privacy of file uploads, avoid query rate limitations, and to ensure full, sensitive analysis of datasets.

While the use of computational tools has increased in humanities research, especially with the growing movement behind the Digital Humanities, there are still widespread barriers regarding the use of newer technologies such as Generative AI. Jaillant and Aske explore some of the obstacles researchers across disciplines face when attempting to incorporate computational methods into their workflow, such as lack of training, lack of access to records, and time constraints (2023). Jaillant and Aske's study is an important consideration for projects whose researchers do not have a strong technical background and are especially impacted by a lack of time and AI-specific training, and who cannot train their own LLM model. This project highlights the use of a widely used, commercially available LLM tool that is expected to minimize such constraints.

IV. Challenges of this dataset

The 1940s and 50s census data provide several unique fields about Japanese and Japanese-American individuals living in Sacramento. Some fields of note include First and Last Names, Sex, Birth Year, Place of Birth, Citizenship, Household ID, and Occupation. An important historical point of note is that due to the internment camps after the 1940s census was taken, many families could not return to their original residences and found lodging elsewhere (Cool 2017), which would result in different Address fields. They may have taken up different jobs and industries, causing differences in the Occupation field. There is also a discrepancy between Japanese given names and English given names; many Japanese-Americans have both, but the order in which they list them or preference for one over the other may change across datasets. This variation in format or values across common fields makes matching individuals a special challenge.

The First and Last Name fields prove to be the largest challenge for this dataset. While individuals with the exact same name recorded in both the 1940s and 50s census would be simple to identify through a direct match search without the need for fuzzy matching, this is not the case for most. This is due to several reasons, including name misspellings or entry errors (e.g. “Hasuo” and “Yasuo”), alternative romanizations for Japanese to English (e.g. “Kikunozo” and “Kikunojo”), the inclusion of initials or middle names (e.g. “William” and “William C.”), and most notably, the alternative order or preference of Japanese and English given names (e.g. “Sakae” and “Jane Sakae”). This latter challenge shows prominently in datasets as a result of post-war pressures faced by many *Nisei* (second generation) citizens who preferred identifying with English names after internment (Densho 2011). Due to differences in how *Nisei* identified, direct matches cannot be made across datasets, and fuzzy matches based on similar name entries or across multiple fields must be introduced in order to confirm a match. While such fuzzy matching has been explored in previous research, LLMs offer a promising alternative due to their analytical power in processing textual data.

V. LLM usage to address challenges

Census data can contain obstacles to research analysis such as interpretation of handwriting, misspellings, or mistakes in data collection. Some of this can be seen with the Sacramento census records, in which some individuals identified through prior manual entity resolution have a slightly misspelled name or different birth years. For example, the birth year for an individual named Kusunosuki Miyai is listed as 1877 in the 1940s dataset and 1878 in the 1950s dataset, but one can be fairly certain that both entries are for the same person due to matches for his name, place of birth, relation, and the names of family members within the same household.

Fields such as Address cannot be relied on individually for matches, as the WWII internment camps greatly displaced Japanese Americans after the war and many were unable to return to their homes. Similarly, Occupation could not be used due to loss of employment during the internment and the need to seek another job afterward. Performing automated entity resolution will need to account for such obstacles, therefore indicating that the best method for doing so would be through fuzzy matching, based on similar data values rather than exact matches.

Lastly, prior manual entity resolution relied heavily on comparing relatives of a potential match across datasets. If a potential match had the same or similar relatives within their household, it was reasonable to confirm the match if other fields also matched. Matching criteria can be broken down into primary field comparisons (First Name, Last Name, Date of Birth, Birth Place, Relative matches) and secondary field comparisons (Occupation, position within household, Marital status), with primary ones holding more assurance of a positive match and secondary ones supporting the confidence of a match. An individual whose name is slightly different, with a date of birth that does not match and a missing place of birth, could still be considered a potential match if other relatives within their household also match across datasets. This kind of matching is difficult to perform with manual coding and time-consuming for manual identification, therefore LLMs may offer an efficient method for doing so.

VI. Conclusion

The data contained in the 1940s and 1950s census collection, while not complicated at first glance, holds deep complexity within the context of the Japanese American World War II internment camps. Cultural influences such as the adoption of English names or name orders, human errors caused by census takers, language-based fluctuations such as alternative romanizations of names, and the historical impact of the camps and ensuing tensions against the Japanese American population resulting in differences in address or occupation all make entity resolution across the 40s and 50s datasets difficult to perform manually by researchers, or through manual coding via Python or SQL. Commercially available LLMs such as ChatGPT offer an opportunity to confront these challenges by performing effective, historically sensitive fuzzy matching of individuals across multiple datasets with consideration for the collection's complexities. While LLM usage must be carefully supervised and manual confirmation of matches may be necessary in working with data in the discussed project, it is expected that integration of LLM into this workflow will help automate more time-consuming and tedious steps of data analysis and pave the way for researchers to computationally examine the historical significance of this collection, such as through data visualizations that may catch patterns yet unexplored.

Bibliography

- Beltran, L., O'Brien, E. P., Jansen, G., & Marciano, R. (2021). A framework for unlocking and linking WWII Japanese American incarceration biographical data. *2021 IEEE International Conference on Big Data (Big Data)*, 2200–2209. <https://doi.org/10.1109/bigdata52589.2021.9671578>.
- Cool, N. (2017, August 3). *Leaving home behind: The fates of Japanese American houses during incarceration* | National Museum of American History. National Museum of American History. <https://americanhistory.si.edu/explore/stories/leaving-home-behind-fates-japanese-american-houses-during-incarceration>
- Densho. (2011, March 16). *Hatsuji becomes Harry: Names and Nisei identity*. Discover Nikkei. <https://discovernikkei.org/en/journal/2011/3/16/hatsuji-becomes-harry/>
- Gnanasekaran, R. K., Perine, L., Conrad, M., & Marciano, R. (n.d.). *Model Selection for HERITAGE-AI: Evaluating LLMs for Contextual Data Analysis of Maryland's Domestic Traffic Ads (1824–1864)*. IEEE Xplore. <https://ieeexplore.ieee.org/document/10825591>.
- Jaillant, L., & Aske, K. (2023). Are users of digital archives ready for the AI era? obstacles to the application of Computational Research Methods and new opportunities. *Journal on Computing and Cultural Heritage*, 16(4), 1–16. <https://doi.org/10.1145/3631125>.
- Jansen, G., & Marciano, R. (2025). Developing computer vision and machine learning strategies to unlock government-created records. *AI & SOCIETY*, 40(6), 4513–4529. <https://doi.org/10.1007/s00146-025-02231-y>.
- National Archives and Records Administration. (n.d.). *Executive order 9066: Resulting in Japanese-American incarceration (1942)*. <https://www.archives.gov/milestone-documents/executive-order-9066>.